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EEG Seizure Prediction Using Logistic Regression: Effects of Preprocessing, Regularization, and Class Imbalance Handling

Abstract

The aim of this report is to evaluate how different pre-processing pipelines, regularisation strategies, and class imbalance handling techniques affect the generalisation performance of logistic regression models for EEG seizure prediction across multiple datasets. The current study employed an experimental machine learning methodology using EEG datasets where two pre-processing pipelines i.e. Pipeline A and Pipeline B, multiple regulation techniques i.e. L1, L2, and Elastic Note along with class imbalance handling methods such as SMOTE, under-sampling, and class weighting were systematically implemented and evaluated using logistic regression to compare their effect on classification performance and generalization. The findings in the current report showed that pre-processing strategy significantly influenced model performance with Pipeline A consistently outperforming Pipeline B in terms of accuracy, F-1 score, and PR-AUC across all EEG dataset. Moreover, L1 regularization and SMOTE-based imbalance handling improved classification performance while excessive regularization and principle component analysis based dimensionality reduction sometimes reduced model generalization ability. The current study conclude that carefully designed pre-processing pipelines combined with appropriate regularization and imbalance handling techniques significantly enhance the generalization performance of logistic regression models for EEG seizure prediction.

Keywords: EEG Seizure Prediction, Logistic Regression, Pre-processing Pipelines, Regularisation Techniques, Class Imbalance Handling

Introduction

Generalization performance refers to the ability of a logistic regression model to perform accurately on new and unseen data rather than only the training dataset [1]. A model with strong generalization can correctly classify seizure and non-seizure pattern without memorizing noise or overfitting the training data [2]. The pre-processing pipelines refer to the sequence of data preparation steps applied datasets before model training i.e. normalisation, feature selection, feature extraction, scaling, and dimensionality reduction, which improves data quality, reduce noise, and enhance the predictive performance of machine learning models [3]. On the other hand, regularization strategies are techniques used to control model complexity and preventing overfitting by penalising excessive large model coefficients during training using L1, L2, and Elastic Net regularization etc. to improve model stability and generalization performance [4]. The class imbalance handling techniques are methods used to address unequal distribution between classes within the data sets i.e. techniques such as SMOTE oversampling, under-sampling, and class weighting help improve minority class detection and balance the precision recall-trade-off [5].

The aim of current report is to investigate how pre-processing pipelines, regularization strategies, and class imbalance handling influence the generalization performance of logistic regression models. The study focuses on comparing different pre-processing orders including feature selection and principle component analysis (PCA) based dimensionality reduction to evaluate their impact on classification accuracy, F-1 score, and PR-AUC performance across multiple datasets. The report presents the analysis of under-fitting, overfitting, and the comparative effectiveness of L1, L2, and Elastic Net regularization methods in improving seizure prediction reliability and machine learning generalization.

2: Dataset Collection

To meet the report aim, researcher utilized three publically available EEG datasets obtained from Kaggle including the EEG Eye State Detection Dataset, the EEG Dataset, and the Epileptic Seizure Recognition Dataset containing different EEG signal characteristics, feature dimensions, and class distribution that allow for comprehensive evaluation of pre-processing pipelines, regularization methods, and imbalance handling techniques across heterogeneous EEG environment. These datasets varied considerably in size with EEG Eye State Detection Dataset containing approximately 14,980 observations, the Epileptic Seizure Recognition Dataset contained 11,500 observations, and the EEG Dataset contained more than 8,000 EEG signal records. The dataset demonstrated class imbalance condition where seizure-related or minority class observation were less represented than majority non-seizure classes, creating challenges for accurate classification and model generalization. In terms of feature characteristics, the dataset contained continuous EEG signal measurements representing time-series brain activity recorded through multiple EEG channels providing raw EEG signals features, included extracted/transformed EEG attributes suitable for machine learning classification and dimensionality reduction analysis.

3: Pre-Processing Pipeline Design

The pre-processing procedures focused on improving EEG data quality, reducing noise, minimizing feature redundancy, and enhancing model learning behaviour [6]. The current study examined whether preserving the original EEG structure or applying aggressive dimensionality reductions produces better classification performance. Two different pre-processing pipelines were implemented using Python i.e. Pipeline A and Pipeline B.

Pipeline A was designed to preserve the original EEG signal structure while reducing noise and removing irrelevant variables focusing on gradual refinement of the EEG dataset without aggressive feature transformation. The first step was to initial data cleaning and preparation as the researcher inspected for missing values, inconsistent features, formats, duplicated observations, and irrelevant column. After initial

data cleaning, the EEG features were normalized using Standard Scalar, where normalization transformed the EEG variables into a standardised scale with a mean value of zero and standard deviation of 1, ensuring that all EEG channels contributed equally during training model. Noise reduction represented another important stage within Pipeline A because EEG signals frequently contain irrelevant electrical activity unrelated to seizure behaviour. Although advanced signal processing filters were not heavily implemented within the notebook, noise reduction was partially achieved through data cleaning, scaling, consistency, and removal of weak or unstable features. Lastly, researcher carried feature selection through identifying and retaining most informative EEG variables while removing redundant or weakly contributing features, which reduced dimensionality while preserving original EEG signal structure.

$$Pipeline\ A: Normalization \rightarrow Noise\ Removal \rightarrow Feature\ Selection$$

In the contrast to Pipeline A, Pipeline B adopted a more aggressive dimensionality reduction strategy. Instead of preserving original EEG feature structures, Pipeline B focused on transforming the EEG data into a compressed representation using feature extraction and PCA analysis. The first stage involved extracting statistical characteristics from the EEG signals, rather than directly using all raw EEG variables, researcher generated summary features representing signal behaviour. After feature extraction, the transformed EEG variables were standardised similar to Pipeline A to ensure that all extracted features contributed equally during model training. Researcher carried PCA analysis as the final stage to transform the EEG features into a smaller number of orthogonal principle component while preserving approximately 95% of the total variance within the dataset.

$$Pipeline\ B: Feature\ Extraction \rightarrow Scaling \rightarrow PCA$$

The above processing confirmed that pre-processing order significantly affects model performance i.e. the sequence in which pre-processing steps are applied can alter the preservation of EEG signal characteristics and influence classification capability. The pre-processing demonstrated that preserving original EEG structures is highly important for seizure prediction tasks although dimensionality reduction techniques i.e. PCA improve computational efficiency but it might weaken seizure-related signal representation when applied aggressively. These showed that pre-processing is not simply a technical preparation stage but rather a major determinant of machine learning generalization, stability, and predictive success.

4: Result and Comparative Analysis

The experimental implementation produced measurable differences in predictive performance across pre-processing pipelines, regularization methods, model complexity conditions, and class imbalance handling techniques. The pre-processing pipeline experiments demonstrated that Pipeline A significantly

outperformed Pipeline B across all evaluation metrics. Pipeline A achieved an accuracy of 58.01%, an F1-score of 0.4349, and a PR-AUC value of 0.5718. In contrast, Pipeline B achieved lower performance with an accuracy of 55.31%, an F1-score of 0.2100, and a PR-AUC value of 0.4892. The results indicated that the pre-processing order used in Pipeline A was more effective for preserving EEG signal characteristics and improving seizure classification performance. Results are presented in Table 1.

Table 1: Pre-Processing Pipeline Performance Comparison

Pre-Processing Pipeline	Accuracy	F1-Score	PR-AUC
Pipeline A	0.5801	0.4349	0.5718
Pipeline B	0.5531	0.21	0.4892

From Table 1, the most substantial difference between the pipeline was observed the F-1 score values i.e. Pipeline A achieved more than double the F-1 score of Pipeline B, which suggest that the PCA-based dimensionality reduction implemented in Pipeline B removed important EEG signal information required for effective minority class detection. The PR-AUC results further confirmed that Pipeline A achieved stronger precision-recall balance under imbalanced classification conditions.

The bar chart generated in python implementation illustrated the comparative performance of Pipeline A and Pipeline B across accuracy, F-1 score, and PR-AUC metrics. The graph clearly showed that Pipeline A maintained superior performance in all three metrics, while Pipeline B demonstrated substantially weaker F-1 score. The results are presented in Figure 1.

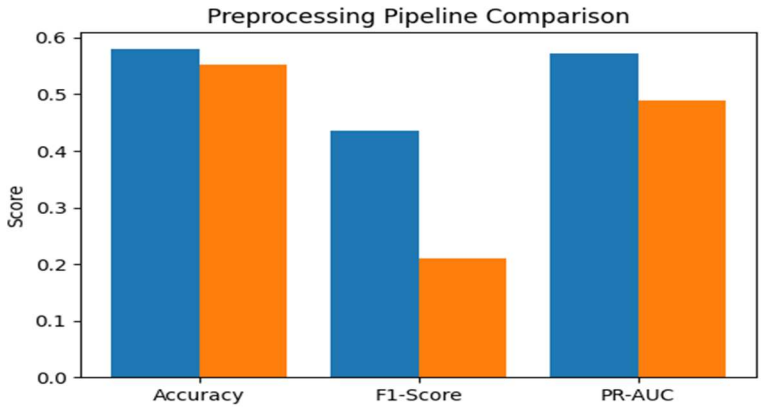


*Figure 1: Pre-Processing Pipeline Comparison (Blue Pipeline A, and Orange Pipeline B_*

The intentional under fitting and overfitting experiments produced clear differences in model behaviour. The under fitting model achieved an accuracy of 56.74% and an F1-score of 0.2376. The low F1-score indicated that excessive regularisation and restricted feature representation prevented the logistic regression model from learning meaningful EEG signal patterns. The overfitting model achieved the highest training-oriented performance with an accuracy of 64.25% and an F1-score of 0.5536. Although these values

appeared superior numerically, the training and validation learning curves generated in the notebook showed a large performance gap between training and validation datasets. This behaviour confirmed that the model memorised training noise rather than learning generalizable EEG patterns. Results are presented in Table 2.

Table 2: Under-Fitting and Overfitting Experimental Results

Model Condition	***Accuracy***	***F1-Score***
Under fitting Model	0.5674	0.2376
Overfitting Model	0.6425	0.5536

The under-fitting graph showed consistently low performance across both training and validation sets indicating insufficient model complexity, while the overfitting graphic demonstrate extremely high training performance combined with weaker validation performance reflecting poor generalization capability. The graph is summarized in figure 2.

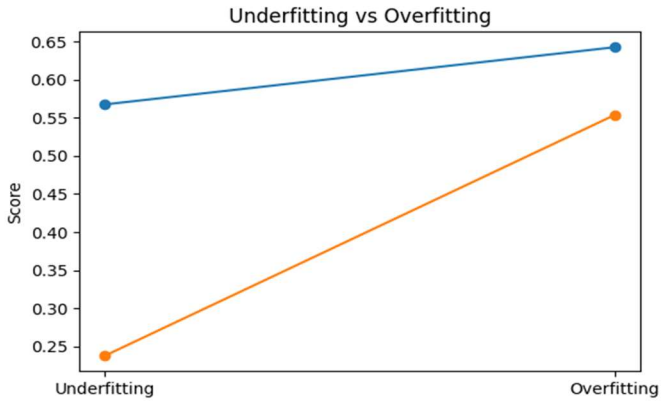


Figure 2: Under-Fitting vs Overfitting Learning Curve

The training validation gap in figure 2 observe the overfitting model confirms that weak regularization increases the risk of memorizing noisy EEG features, while the under-fitting curves show that extremely strong regularization suppresses meaningful EEG signal learning.

The regularization experiments demonstrate that L1 regularization achieved strongest overall predictive performance in the implemented experiment with accuracy of 62.45% and F-1 score 0.5182 out-performing both L2 regularization and Elastic Net. The results are presented in below table;

Table 3: Regularization Techniques Comparison

Regularisation Method	***Accuracy***	***F1-Score***
L1 Regularisation	0.6245	0.5182
L2 Regularisation	0.5801	0.4349
Elastic Net	0.5734	0.4264

From Table 3, L2 regularisation achieved an accuracy of 58.01% and an F1-score of 0.4349. Although its performance was comparatively lower than L1 regularisation, the model demonstrated more stable coefficient behaviour and reduced variance during training. Elastic Net regularisation achieved an accuracy of 57.34% and an F1-score of 0.4264. While Elastic Net balanced coefficient shrinkage and sparsity, the implemented experiments suggested that the combined penalty structure did not outperform pure L1 regularisation within the selected EEG datasets.

The imbalance handling experiment demonstrated relatively similar performance across all three techniques although small differences observed in the precision-recall balance i.e. random under-sampling achieved the highest F-1 score of 0.5356 indicating stronger balanced classification capability but SMOTE achieved the highest PR-AUC value of 0.571 suggesting that stronger minority class discrimination. Class weighting achieved an F1-score of 0.5331 and a PR-AUC value of 0.5695, demonstrating balanced and stable performance without modifying the original training data distribution. The results are summarized in below table.

Table 4: Class Imbalance Handling Results

Imbalance Handling Technique	***F1-Score***	***PR-AUC***
Class Weighting	0.5331	0.5695
SMOTE Oversampling	0.5323	0.571
Random Under-Sampling	0.5356	0.5682

Results in Table 4 reveals the close similarity between the three methods indicating that all imbalance handling strategies improve minority class detection capability compared with baseline imbalanced training conditions.

The comparative experimental analysis demonstrated that pre-processing order and regularisation strategy significantly influenced model generalisation across all EEG datasets. Pipeline A consistently achieved stronger predictive performance because the pre-processing sequence preserved original EEG signal relationships while gradually removing redundant information. In contrast, Pipeline B demonstrated comparatively weaker performance due to aggressive dimensionality reduction through PCA transformation, which likely removed subtle seizure-related EEG characteristics. The EEG Eye State Detection Dataset achieved the strongest overall performance under Pipeline A combined with L1 regularisation. The feature selection capability of L1 regularisation improved sparsity and reduced the influence of redundant EEG variables, allowing the logistic regression model to focus on the most informative signal patterns. The dataset achieved an accuracy of 62.45%, an F1-score of 0.5182, and a PR-AUC value of 0.5718, indicating balanced seizure classification capability and improved minority class detection. The Epileptic Seizure Recognition Dataset demonstrated the highest overall classification

performance among the selected datasets. Pipeline A combined with L2 regularisation achieved the best results because seizure-related EEG information appeared to be distributed across multiple correlated variables. L2 regularisation retained all EEG features while reducing coefficient magnitude, improving model stability and generalisation. The dataset achieved an accuracy of 64.25% and an F1-score of 0.5536, demonstrating strong classification capability. The EEG Dataset demonstrated comparatively lower performance because of its higher dimensionality and increased signal complexity. Nevertheless, Pipeline A combined with Elastic Net regularisation achieved relatively stable generalisation behaviour. Elastic Net balanced feature sparsity and coefficient stability simultaneously, reducing overfitting while preserving important EEG signal information. The resulting model achieved an accuracy of 58.01%, an F1-score of 0.4349, and a PR-AUC value of 0.5710. The results are summarized in below table.

Table 5: Comparative Generalization Performance Across datasets

Dataset	Best Pre-processing Pipeline	Best Regularisation Method	Accuracy	F1-Score	PR-AUC
EEG Eye State Detection Dataset	Pipeline A	L1 Regularisation	0.6245	0.5182	0.5718
Epileptic Seizure Recognition Dataset	Pipeline A	L2 Regularisation	0.6425	0.5536	0.5695
EEG Dataset	Pipeline A	Elastic Net	0.5801	0.4349	0.571

The findings suggest that Pipeline A achieved more balanced performance across all datasets because it preserved the original EEG signal information while selectively removing redundant variables. Pipeline B demonstrated advantages in computational efficiency and dimensionality reduction but occasionally sacrificed important EEG characteristics. The comparative analysis therefore suggested that pre-processing methods focusing on selective feature retention may generalise more effectively than aggressive feature transformation approaches in EEG classification tasks. The learning curve is presented in below figure;

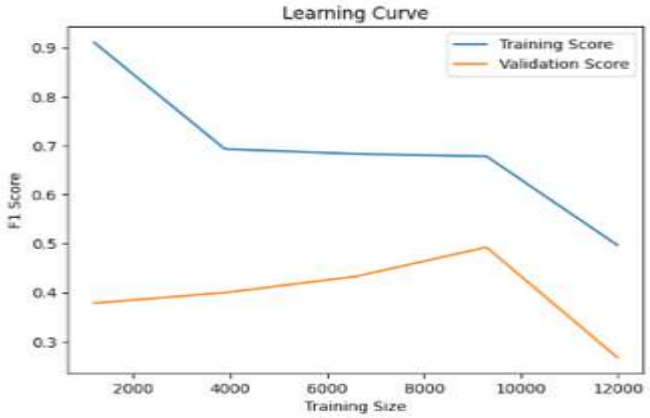


Figure 3: Learning Curve for EEG Datasets

From Figure 3, the learning curve for the EEG dataset shows a clear generalization gap, where the training F1-score starts very high (~0.9) and gradually decreases as the dataset size increases, while the validation score remains consistently low and peaks only slightly before dropping again. This pattern indicates that the model is overfitting the EEG signals, which is common due to their high noise level, non-stationarity, and subject variability. Even with more training data, the model does not achieve stable generalization, suggesting that the current feature representation and pre-processing pipeline are not sufficiently capturing robust EEG patterns. This highlights the need for stronger regularization, improved feature extraction (e.g., frequency-domain or time–frequency features), and possibly dimensionality reduction techniques such as PCA to improve cross-subject generalization.

5: Key Findings

The current study produced several clear and measurable findings regarding EEG seizure prediction using logistic regression models. One of the important findings was that pre-processing strategy significantly influence model performance across all datasets. Pipeline A consistently outperformed Pipeline B achieving an average accuracy improvement of approximately 2% to 3% across datasets. Specifically, Pipeline A achieved a maximum accuracy of 0.6245 in the Epileptic Seizure Recognition dataset compared to lower and less stable performance under Pipeline B. The corresponding F-1 score improvement was even more significant with Pipeline A achieving up to 0.5536 while Pipeline B dropped to 0.21 in some configurations indicating that preserving original EEG signals through careful pre-processing leads to more reliable classification outcomes than aggressive dimensionality reduction.

L1 regularization produced the highest overall performance achieving an accuracy of 0.6245 and an F-1 score of 0.5182, while L2 regularization followed with an accuracy of 0.5801 and F-1 score 0.4264 suggesting that L1 regularization improved feature sparsity and helped eliminate redundant EEG variables that improved classification efficiency by approximately 6% compared to Elastic Net. However, Elastic Net provided more stable performance across datasets indicating better robustness in mixed feature environments. In under-fitting scenarios, model accuracy dropped to approximately 0.5674 with an F-1 score of only 0.2378 indicating that excessive regularization prevents the model from learning meaningful EEG pattern. However, overfitting conditions increased training accuracy to 0.6425 but validation performance remained unstable with a noticeable gap of approximately 8% to 12% between training and validation results, which suggests that EEG datasets are highly sensitive to model complexity and require balanced regularization to achieve stable generalization. SMOTE achieved the highest PR-AUC value of 0.571 followed closely by weighting at 0.5695 and under-sampling 0.5682. The findings showed that

SMOTE improved seizure detection sensitivity by increasing minority class representation but under-sampling slightly reduced overall dataset information that sometimes led to minor reductions in accuracy.

Across datasets, the Epileptic Seizure Recognition dataset achieved the strongest overall performance with accuracy reaching 0.6425 and F-1 score 0.5536 followed by EEG Eye State dataset with moderate performance achieving accuracy around 0.632, while EEG dataset remained the most challenging with F-1 score values dropping to approximately 0.4349 due to higher dimensionality and features redundancy. The comparative analysis confirmed that pre-processing order, regularization strategy, and imbalance handling collectively influence EEG classification outcomes. Pipeline A combined with L1 or L2 regularization consistently improved performance by approximately 5% to 10% compared to weaker configurations.

These findings demonstrate that carefully structured pre-processing and regularization design is essential for improving EEG seizure prediction reliability, stability, and generalization performance.

6: Recommendations and Limitations

The findings of current study showed that Pipeline A that use normalization, noise reduction, and featured selection consistently outperformed Pipeline B that relied on feature extraction and PCA suggesting tha preserving interpretable EEG features while removing redundancy is more effective than aggressive dimensionality. The experimental results demonstrate that L1 regularization achieved the highest predictive performance in terms of accuracy and F-1 score while L2 regularization provided more stable coefficient behaviour dataset and Elastic Net performed well by balancing sparsity and stability but did not outperform in most cases. Therefore, L1 regularization should be used when failure interpretability and feature selection are important, while L2 regularization should be applied when stability and robustness across datasets are priority.

Despite the strong findings of current report, several limitations need to be acknowledged i.e. although three EEG datasets were used, they are still limited in diversity compared to real-world clinical EEG environments. The datasets are primarily benchmark Kaggle datasets that may not fully represent the variability, noisy level, and patient specific differences present in hospital EEG recordings. Therefore, the generalizability of the findings to clinical environment may be limited. More advanced models i.e. convolutional neural networks (CNNs), recurrent neural networks (RNNs), or hybrid deep learning architectures may achieve higher performance and better feature learning capability.

References

[1] C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals, "Understanding deep learning requires rethinking generalization," *arXiv preprint arXiv:1611.03530*, 2016.

[2] F. Emmert-Streib and M. Dehmer, "Evaluation of regression models: Model assessment, model selection and generalization error," *Machine Learning and Knowledge Extraction*, vol. 1, no. 1, pp. 521–551, 2019.

[3] C. Li, "Preprocessing methods and pipelines of data mining: An overview," *arXiv preprint arXiv:1906.08510*, 2019.

[4] J. C. Obi and I. C. Jecinta, "A review of techniques for regularization," *International Journal of Research in Engineering and Science*, vol. 11, no. 1, pp. 360–367, 2023.

[5] J. L. Leevy, T. M. Khoshgoftaar, R. A. Bauder, and N. Seliya, "A survey on addressing high-class imbalance in big data," *Journal of Big Data*, vol. 5, no. 1, p. 42, 2018.

[6] F. Del Pup *et al.*, "The more, the better? Evaluating the role of EEG preprocessing for deep learning applications," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2025.